SHIMIZU CORPORATION

CONSTRUCTION OF A PROPOSED DEVELOPMENT AT NEIL ROAD, SINGAPORE
DNR Project
BILLS OF QUANTITIES

S/No	Description	Unit	Blk S	Blk M	Total	Rate	Amount
	Installation on site: Mar 2023 (Post-installation) Installation on site: Nov 2022 (Cast-in installation)						
	Security Bollard						
	Supply, fabricate, and install shallow **removable** Stainless steel 304 with hairline finish bollard **[Type B2]** and all necessary connection and accessories: all in accordance with the Specification and as shown on the Drawings: to the approval of the SO and/or Contractor. Refer to Spec: TS17/13 - crash-rated shallow bollards shall be between 250-350mm diameter, 900-1100mm ht w/ min rating of PAS 68:2013 standard of of 7500kg at 48km/h, with Engineering claculation						
a	Basement 1, STUDIO E/P5887/TD/AD/S/FP/02/C3 Rev C	no	6	-	6	$ 3,150.00	$ 18,900.00
b	Stainless steel sleeve for crash bollard	no	6	-	6	$ 750.00	$ 4,500.00
	Supply, fabricate, and install fixed bollard **[Type B3]** and all necessary connection and accessories: all in accordance with the Specification and as shown on the Drawings: to the approval of the SO and/or Contractor. Refer to Spec: TS17/13 - crash-rated fixed bollards shall be between 250-350mm diameter, 900-1100mm ht w/ min rating of PAS 68:2013 standard of of 7500kg at 48km/h, compolsory with test report						
c	Level 1, STUDIO E/P5887/TD/AD/S/FP/03/C3 Rev C	no	8		8	$ 2,900.00	$ 23,200.00
d	Level 1, STUDIO E/P5887/TD/AD/EW/01/C6 Rev	no		2	2	$ 2,900.00	$ 5,800.00
e	'Level 1, STUDIO E/P5887/TD/AD/EW/02/C6 Rev	no		51	51	$ 2,900.00	$ 147,900.00
f	Level 1 outside boundary nearby by Neil Road, STUDIO E/P5887/TD/AD/S/FP/03/C3 Rev C	no	5		5	$ 2,900.00	$ 14,500.00
g	Level 1 outside boundary gridline 9/N, STUDIO E/P5887/TD/AD/S/FP/03/C3 Rev C (SOI)	no	6		6	$ 2,900.00	$ 17,400.00
h	Level 1 outside boundary gridline 1/N, STUDIO E/P5887/TD/AD/S/FP/03/C3 Rev C (SOI)	no	7		7	$ 2,900.00	$ 20,300.00
i	Blk S Level 1 bebside Staircase 1 , under SOI-A-011	no		4	4	$ 2,900.00	$ 11,600.00
j	Sentry 1 @ Level 1, under **NEW SOI** (Top-down slab Construction)	no		7	7	$ 2,900.00	$ 20,300.00
k	Level 1, STUDIO E/P5887/TD/AD/M/FP/04/C5 Rev C	no		3	3	$ 2,900.00	$ 8,700.00
l	Stainless steel sleeve for crash bollard	no	26	67	93	$ 750.00	$ 69,750.00
	Supply, fabricate, and install shallow **fixed** Stainless steel 304 with hairline finish bollard **[Type B4]** and all necessary connection and accessories: all in accordance with the Specification and as shown on the Drawings: to the approval of the SO and/or Contractor. Refer to Spec: TS17/13 - crash-rated shallow bollards shall be between 250-350mm diameter, 900-1100mm ht w/ min rating of PAS 68:2013 standard of of 7500kg at 48km/h, with Engineering claculation						
m	Basement 1, refer to Drawing no: STUDIO E/P5887/TD/AD/S/FP/02/C3 Rev C	no	3		3	$ 2,900.00	$ 8,700.00
n	Basement 1, refer to STUDIO E/P5887/TD/AD/M/FP/03/C3 Rev B	no		1	1	$ 3,500.00	$ 3,500.00
o	Basement 2, refer to STUDIO E/P5887/TD/AD/M/FP/02/C3 Rev B	no		2	2	$ 3,500.00	$ 7,000.00
p	Stainless steel sleeve for crash bollard	no	3	3	6	$ 750.00	$ 4,500.00
	Supply, fabricate, and install safety bollard **[Type B5]** and all necessary connection and accessories: all in accordance with the LTA's standard and as shown on the Drawings: to the approval of the SO and/or Contractor. Refer to Spec: TS17/13 - safety bollards shall be constructed and finishes in accordance to LTA's standard detail on safety bollard at Bus Stop - posted speed: 60km/h & 70km/h. **Sub-contractor to provide calculation.**						
	(Drawing ref: LTA SDRE14/5/RAL1-TYPE A for details)						
q	Level 1, STUDIO E/P5887/TD/AD/EW/02/C6 Rev	no		11	11	$ 750.00	$ 8,250.00
r	Level 1, under **SOI-A-0018**	no		12	12	$ 750.00	$ 9,000.00
	Others:						
s	Preliminary	LS			1	$ 15,000.00	$ 15,000.00
t	PE endorsement fee & Supervision	LS			1	$ 4,500.00	$ 4,500.00
u	Goodwill Discount	LS			1	$ (3,300.00)	$ (3,300.00)
						TOTAL AMOUNT	**420,000.00**